1.CI/CD Pipeline Integration

- Cloud Build, GitHub Actions, GitLab CI for GKE
- Automated testing and deployment workflows
- Multi-stage pipelines with approvals
- Rollback strategies and blue-green deployments

Hands On: CI/CD pipeline to push images to Artifact Registry and deploy to GKE

1.CI/CD Pipeline Integration

CI/CD (Continuous Integration / Continuous Deployment) is a set of practices that automate code integration, testing, and deployment, ensuring faster delivery with minimal errors.

Application development process

1. Monolith Pipeline Pushback Approach

Single pipeline handles all services Always downgraded Hard when teams want phases Hard to complex/maintains Pre-test logics

2. Stages / Pipelines Approach (Microservices Approach)

Pipeline
Pipeline
Pipeline